



UMEÅ UNIVERSITY

A Formula for Investing in the Semiconductor Industry

**An Empirical Study Using Firm Fundamentals
and System GMM**

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Abstract

This thesis investigates whether firm-specific financial metrics can predict stock returns in the semiconductor industry and whether these predictors can form the basis of a systematic investment strategy. Using a panel of fifty global semiconductor firms from 2006 to 2022, the analysis applies a two-step System Generalized Method of Moments estimator to address endogeneity and firm heterogeneity. Return on assets emerges as the most robust and statistically significant predictor of future stock returns, reflecting the central role of asset efficiency in this capital-intensive sector.

Portfolio tests compare buy and hold and annually rebalanced strategies against the NASDAQ benchmark. The results show that ROA based portfolios achieve superior raw performance, particularly when rebalanced annually, although excess returns largely reflect higher market exposure once risk adjustment is applied. Within the framework of the Adaptive Market Hypothesis, the findings suggest that profitability-based strategies may occasionally capture time varying inefficiencies, but their effectiveness depends on evolving market conditions.

The results overall show that, in a semiconductor-specific dynamic panel that accounts for unobserved firm effects and endogeneity, return on asset is the only robust fundamental predictor of stock returns. This motivates an investment formula that can narrow the set of firms investors need to evaluate when making investment decisions in this sector.

Keywords: Semiconductor industry, firm fundamentals, investment strategy, dynamic panel data, System GMM, capital asset pricing model, adaptive market hypothesis, risk-adjusted returns, fundamental analysis.

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1 Introduction

1.1 Background

The semiconductor industry has experienced rapid growth in recent years, driven largely by technological advancements and the increased demand for high-performance computing. This trend has been further accelerated by the expansion of artificial intelligence (AI). As a capital- and knowledge-intensive sector, semiconductors remain central to technological development, but the rapid pace of change also intensified competition within the industry. According to the Semiconductor Industry Association (2025), global semiconductor sales increased by 19.1% in 2024, and double-digit growth is projected to continue in 2025. The ten largest firms accounted for 67% of global market share in 2024 (Gordon 2025), with Nvidia leading in growth and value (Erricker 2025). McKinsey & Company (2025), one of the world's largest management consulting firms, reports that in 2024 the top 5 percent of semiconductor firms generated nearly all of the industry's economic profit, while returns among the remaining 95 percent declined.

Despite the strong performance, McKinsey's findings show that only a small fraction of firms capture most industry profit, while most struggle to generate returns. For individual investors, this concentration makes stock selection more difficult if you want to beat the market. At the same time, cyclical nature of the industry and the technical expertise often required to understand the business and assess firm performance further complicate investment decisions. While investment strategies are common, most are either highly complex or inaccessible to non-professional investors, leaving few simple and transparent methods available.

This thesis examines whether firm-level financial metrics can predict stock performance in the semiconductor industry. Prior research suggests that profitability, growth, and innovation may predict returns, but results vary across the literature and often rely on empirical approaches that do not jointly account for firm-specific heterogeneity, return persistence, and endogeneity between returns and firm fundamentals. This thesis contributes by providing semiconductor-specific evidence on how commonly used fundamentals should be interpreted as return predictors once firm effects, return persistence, and endogeneity are accounted for. Using the two-step System GMM estimator, we can obtain consistent estimates in a setting where returns and fundamentals may influence each other over time. The estimates are then used to develop a simple and transparent investment formula that can be used for screening to reduce the set of firms investors need to evaluate.

The purpose of this thesis is to develop a fundamentals-based investment formula that is accessible and replicable for individual investors. Using a panel of fifty listed semiconductor

firms from 2006 to 2022, System GMM is used to identify which financial metrics predict stock returns.

1.2 Research Question

1. Which firm-specific financial metrics, identified through System GMM estimation, consistently predict stock returns in the semiconductor industry?
2. Can the identified financial metrics be transformed into a transparent and systematic investment rule that delivers superior performance relative to market and sector benchmarks?
3. Does the fundamentals-based strategy generate excess returns beyond those explained by risk exposure and market movements?

1.3 Delimitation

This thesis focuses exclusively on firms operating within the semiconductor industry, as defined by the *Global Industry Classification Standard* (GICS) maintained by MSCI and S&P global. The GICS framework assigns each publicly traded firm a unique industry classification based on its main business activity. The relevant classification for this study is the *Semiconductors & Semiconductor Equipment* industry group (GICS code 4530), which is a sub-section of the broader *Information Technology* sector (code 45) (GICS® - Global Industry Classification Standard n.d.).

To clearly define the sample, this thesis follows a common classification used by financial data and consists of four main categories (Hurtarte, Wolfsheimer, and Tafoya 2007)

1. *Integrated Device Manufacture* firms (IDM:s) build and manage their own manufacturing facilities, which lets them directly control production capacity, design, process steps and resources.
2. *Fabless* firms handle only the design of chips and rely entirely on external foundries for manufacturing.
3. *Foundries* focus their entire business on manufacturing chips for fabless firms and do not engage in designing or marketing their own products.
4. *Semiconductor Equipment Manufacturers* develop and supply the tools and machines used in chip production process.

2 Previous research

2.1 Adaptive Market Hypothesis and Empirical Investment Strategies

Empirical studies support time-varying efficiency, suggested by the Adaptive Market Hypothesis (AMH). Cruz-Hernández and Mora-Velencia (2024) found that return predictability changes over time, with markets switching between periods of efficiency and inefficiency. A study on the Dhaka Stock Exchange (1995 to 2018) shows that momentum and contrarian profits fluctuate with market conditions. Medium-term momentum and long-term reversals are more evident during normal or bullish periods but disappear under crashes and bubbles (Akhter and Yong 2019). While informative, such results are limited to emerging markets and may not translate directly to highly globalized, capital-intensive industries such as semiconductors.

Evidence from the quality-minus-junk (QMJ) factor shows that high-quality stocks, defined by profitability, growth, and safety, earn robust returns across countries (Asness, Frazzini, and Pedersen 2019). However, the price of quality changes over time, indicating that the strength of fundamentals as predictors is not constant but conditional. This aligns with the AMH. Taken together, these studies suggest that while fundamental signals exist, their predictive power varies with market regimes. What remains unclear is whether such time-varying predictability also characterizes industries with pronounced structural cycles, such as semiconductors.

2.2 Performance Indicators and Empirical Methods

Research on the relationship between firm fundamentals and stock performance uses a range of empirical approaches, which means results are not always directly comparable across studies. Gschwendtner and Hauser (2016) analyse the relationship between profit persistence and factor-adjusted stock returns using firm-level regressions with lagged explanatory variables. While they incorporate persistence measures based on how profit evolve over time within each firm their return equation remains static and does not model the feedback between returns and firm fundamentals. Baker, Stein and Wurgler (2003) show that market valuation can influence firms' financing and investment decisions, which can in turn affect future fundamentals, motivating a potential two-way feedback between fundamentals and returns. Nevertheless, their results show that firms with more persistent operating performance tend to exhibit different stock returns, indicating that profitability contains information relevant for explaining return variation. In turn, this motivates a dynamic return specification in which stock returns evolve over time and firm fundamentals are treated as potentially endogenous, rather than assuming a one-way relationship.

Portfolio-based methods are also widely applied to study the relationship between firm characteristics and stock returns. Asness et al. (2019) construct a quality-minus-junk factor using long-short portfolios based on profitability, growth, and safety, where profitability is captured by measures such as return on equity (ROE) and return on asset (ROA), and evaluate performance through portfolio returns and factor-model alphas. Ghazal et al. (2024) similarly rely on Fama-French portfolio sorts, ranking firms by R&D intensity and assessing abnormal performance using Carhart four-factor alphas. While these approaches present systematic return differences across groups of firms, they operate at the portfolio level and do not estimate firm-specific predictive relationships, persistent returns, or address potential endogeneity between fundamentals and stock returns.

Some studies use a dynamic panel estimator to examine the relationship between firm characteristics and performance. Alexakis, Patra, and Poshakwale (2010) estimate a dynamic panel model of stock returns using the Generalised Method of Moments (GMM) estimator, which included lagged returns and treated accounting ratios as potentially endogenous. Their approach allows stock returns to depend on past returns and on firm fundamentals and shows that profitability and liquidity ratios help explain stock returns. Tsung-chun et al. (2019) also used System GMM to analyse the effects of R&D intensity on firm performance in the Taiwanese semiconductor industry. They modelled current ROA as a function of past ROA, R&D intensity (including lagged effects), firm size, leverage, and asset growth. Together, these studies provide a methodological foundation for applying the GMM to firm-level analysis, which this study extends to stock return predictability in the semiconductor industry.

In summary, existing studies agree that firm fundamentals such as profitability, growth, margins, and R&D are related to stock returns. However, the empirical methods differ, ranging from firm-level regressions with static return specifications, to portfolio-based methods that assess returns across groups of firms, and to dynamic panel GMM models that allow returns or firm performance to depend on past performance and treat firm fundamentals as potentially endogenous. These differences help explain why some studies put emphasis on ROE or ROA and other R&D or revenue growth. By taking all these different metrics and applying a dynamic model, it becomes possible to more clearly assess which firm characteristics are informative for predicting stock returns.

2.3 Industry-Specific Determinants of Performance

The semiconductor industry is capital and R&D intensive, with firms typically spending 15-20 percent of revenues on R&D (Statistics (NCSES) 2024). These investments contribute to large

fluctuations in financial performance. Ehm and Ponsignon (2012) show that revenues often change by 20-30 percent annually despite stable macroeconomic conditions, driven by inventory and capacity cycles. Given that these fluctuations originate within the industry, and not from the broader market, it strengthens the case for comparing firms relative to each other instead of relying solely on macro-based predictors. In turn it supports the use of a firm-level panel where time effects absorb common shocks, allowing the analysis to focus on differences in fundamentals across firms.

Furthermore, Trucmel (2023) found that semiconductor stocks differ noticeably in their return patterns and volatility across firms, even under similar market conditions. The study also reports periods where semiconductor stocks move differently from the market index during supply chain disruptions or shifts in demand. These results are consistent with the sector's cyclical behaviour and indicate that performance can be driven by firm-specific adjustments in capacity, inventories, and R&D. In turn, supports the idea of creating an empirical model which contains several profitability and innovation indicators.

Qiao and Lu (2024) compare integrated device manufacturers (IDMs) with fabless firms and show that two groups differ noticeably in how they respond to changes in demand. IDMs benefit from scale and internal fabrication but carry higher fixed costs, which makes them more exposed to downturns. Fabless firms have lighter asset structures and adjust more easily. However, they do not capture the full upside during periods of expansive growth. Therefore, these distinctions are directly relevant for this thesis, since includes both IDMs and fabless firms. The variation in business models implies that profitability metrics such as ROA and ROE may not have the same interpretation across firm types. Asset efficiency is more important for IDMs, while flexibility and innovation matter more for fabless firms. It strengthens the idea to test several indicators rather than relying on a single measure and highlights the importance of finding a metric that can capture both types effectively.

2.4 Research Gap and Motivation

The literature shows that profitability, margins, growth, and R&D can predict stock returns, but findings differ across studies because they rely on different empirical approaches. Firm-level regressions, portfolio sorts, and models that do not jointly account for dynamics and endogeneity each capture parts of the relationship between fundamentals and returns, but do not fully address the possibility that firm characteristics and past returns influence each other over time (Asness, Frazzini, and Pedersen 2019; Ghazal et al. 2024; Gschwandtner and Hauser

2016). As a result, the relative importance of indicators such as ROA, ROE, or R&D intensity remains uncertain.

Research in the semiconductor industry highlights the role of R&D, asset intensity, and capacity cycles, yet few studies identify which single firm-level metric provides the most robust relationship with future stock returns. Given the industry's volatility and structural differences, prior work has largely focused general financial performance, not stock return predictability. There is also limited evidence on how fundamentals can be used to form a simple and transparent investment rule in the semiconductor industry.

A methodological gap follows from these limitations. The relationship between past returns and firm fundamentals creates endogeneity that cross-sectional and static models cannot address. A dynamic panel estimator is needed to control for firm-specific effects and to obtain consistent estimates when returns and fundamentals influence each other over time. System GMM is well suited for this purpose because it accounts for firm-specific heterogeneity, lagged dependence in returns, and endogeneity in the independent variables.

System GMM is an established method, and this thesis applies it to semiconductor firms to identify which fundamentals have predictive value. The approach helps to handle differences across firms and changes over time. The investment rule that follows also makes the analysis more practical by reducing the number of firms an investor needs to review. It provides a straightforward way to focus on the most relevant indicators, which can support investors who have less experience or limited time.

3 Theory

The design of an empirical investment strategy requires a clear theoretical foundation that links firms' fundamentals to stock performance. This chapter develops the framework that guides both the selection of explanatory variables and the evaluation of portfolio outcomes. The first section discusses the Adaptive Market Hypothesis (AMH), which provides a behavioural alternative to the Efficient Market Hypothesis (EMH) and explains why the predictive power of fundamentals may vary over time. The second section presents the Capital Asset Pricing Model (CAPM), which explains the relationship between expected returns and systematic risk and provides a way of assessing whether a fundamentals-based strategy generates excess returns beyond market exposure. The final section outlines the firm-level financial metrics most often associated with persistent profitability, thereby motivating their inclusion in the empirical analysis of semiconductor firms.

3.1 Adaptive Market Hypothesis

The Adaptive Markets Hypothesis (AMH), developed by Andrew W. Lo (2004) and grounded in Simon's (1955) concept of bounded rationality, views financial markets as environments where investors adjust their behaviour over time rather than always acting rationally. Lo (2004) summarises this framework through five principles: (i) individuals act to survive and adapt rather than to maximise utility optimally; (ii) market dynamics are driven by competition among participants; (iii) adaptation occurs through heuristics and feedback learning, so decision rules are selected and updated through experience rather than derived from utility maximisation; (iv) market efficiency is not fixed but evolves over time; and (v) financial innovation and external changes continually reshape market outcomes. Under this view, *loss aversion*, *overconfidence*, and *overreaction* are not anomalies but adaptive responses that emerge from changing incentives and competitive pressures.

Overconfidence means that investors place too much weight on their own signals because they overestimate the precision of their private information. Daniel, Hirshleifer, and Subrahmanyam (1998) develop this idea and argue that prices can overreact to private signals and adjust only gradually as further information gets priced in, which creates patterns of under- and overreactions over time. Loss aversion means that people are more averse to losses than they are attracted to gains. In other words, losses are weighted more heavily than equal-sized gains (Tversky and Kahneman 1991); for example, losing €100 requires a gain of more than €100 for an individual to feel equally well off.

In the empirical model, these behavioural patterns are not measured directly, but they affect how quickly prices incorporate information. If investors initially react too little or too much to news, prices may adjust gradually over several periods rather than fully within one period. This is why past returns can contain information about current returns. The specification does not identify loss aversion or overconfidence; instead, these mechanisms are treated as potential sources of the frictions that make short-horizon predictability plausible. Furthermore, if prices incorporate firm-level information only gradually, lagged fundamentals may also contain information about current returns. Therefore, the model in Chapter 4 also includes lagged financial metrics from firms' annual reports.

The AMH also provides a rationale for using implementing rule-based decision-making. If investors operate with heuristics that evolve through experience, a systematic selection rule based on stable fundamentals can reduce the influence of short-term behavioural responses. In

this thesis, the investment rule and annual rebalancing are designed to apply the same criteria each year, supporting consistency and limiting the impact of short-term decision-making.

Lastly, the AMH helps interpret the empirical results. If market efficiency varies over time, returns may not be fully explained by models such as the Capital Asset Pricing Model (CAPM) (Sharpe 1964) in all periods. When markets adjust slowly, it is plausible that fundamentals can help predict returns because prices do not incorporate new information immediately. The AMH does not determine the choice of System GMM, but it supports the use of lagged variables and provides a theoretical basis for studying short-term predictability.

3.2 Capital Asset Pricing Model

The Capital Asset Pricing Model (CAPM) introduced by Sharpe (1964), proposes a linear relationship between the expected return of an asset and systematic market risk. It assumes that in equilibrium only non-diversifiable risk is priced, while firm-specific risk can be eliminated through diversification. The expected return on an asset is expressed as

$$E(R_i) = R_f + [E(R_M) - R_f]\beta_{iM} \quad i = 1, \dots, N.$$

where $E(R_i)$ is the expected return on asset i , R_f is the risk-free rate, $E(R_M)$ is the expected return on the market portfolio, and β_{iM} measures the asset's exposure to systematic risk, defined as the covariance between the asset and the market divided by the variance of the market return. A beta of one implies the asset moves with the market, while values above or below indicate greater or smaller sensitivity to market movements.

Systematic risk reflects economy-wide sources of uncertainty such as interest rate changes, inflation, or geopolitical events (Fama and French 2004). These affect all securities and cannot be diversified away, unlike unsystematic firm-specific shocks.

To evaluate whether a strategy generates returns beyond those attributable to systematic risk, Jensen's alpha is estimated through time-series regression (Jensen 1968).

$$R_{it} - R_{ft} = \alpha_i + \beta_{iM}(R_{Mt} - R_{ft}) + \varepsilon_{it}$$

Here, α_i captures abnormal performance unexplained by exposure to the market. In equilibrium, CAPM predicts that alpha should be zero. A positive and statistically significant alpha indicates excess returns that cannot be attributed to risk exposure alone, potentially reflecting mispricing or omitted risk factors.

In this thesis, CAPM serves as the benchmark for testing whether the investment formula delivers genuine excess returns beyond market movements, directly addressing the third research question.

3.3 Fundamental Determinants of Stock Performance

A large body of financial theory and empirical research emphasizes firm fundamentals as key drivers of long-term stock performance. Graham and Dodd (2023) argue that persistent profitability reflects durable competitive advantage, enabling firms to sustain above-average returns even in competitive markets. Building on this idea, financial metrics serve not only as accounting measures but also as indicators of a firm's underlying economic position, efficiency, and resilience. Calculations are presented in the Appendix (A2.1 – A2.7).

3.3.1 Profitability measures.

Return on Equity (ROE) captures how efficiently a firm generates profits relative to shareholder capital, while Return on Asset (ROA) evaluates the efficiency with which a firm utilizes its total asset base. Return on Capital Employed (ROCE) complements these measures by focusing on earnings generated relative to the total capital invested. Persistent profitability in these ratios signals the presence of structural advantages such as cost leadership, technological edge, or market power. In the semiconductor industry, which is highly capital intensive, ROA reflects a firm's ability to operate costly fabrication facilities efficiently, making it a theoretically strong predictor of performance.

3.3.2 Operational efficiency

Operating margin measures profitability from core activities relative to sales, indicating whether a firm can maintain strong pricing power and cost control. High and stable operating margins suggest product differentiation or supply chain advantages. In semiconductors, where margins are sensitive to capacity utilization and cyclical demand, operating profitability is a critical indicator of competitive positioning.

3.3.3 Leverage

The debt-to-equity ratio (D/E) captures the extent to which a firm relies on debt financing. Moderate leverage can improve returns by boosting profits, but excessive leverage increases financial vulnerability, especially in cyclical industries. Given the capital intensity of semiconductor firms, where large investment in fabrication plants and equipment is often debt-financed, leverage directly affects risk exposure and the sustainability of profitability.

3.3.4 Innovation and growth

R&D intensity, measured as R&D expenditures relative to sales, reflects a firm's commitment to innovation and technological leadership. High R&D spending may depress short-term profitability but is often critical for sustaining competitive advantage in the long term. Semiconductor firms routinely allocate 15 to 20 percent of revenues to R&D, making this metric particularly relevant for explaining stock performance. Revenue growth complements this perspective by capturing expansion in market share and demand. In rapidly evolving technology markets, consistent revenue growth signals adaptability and market acceptance of innovations.

Taken together, these indicators provide a comprehensive view of profitability, efficiency, risk and growth. In the context of this thesis, they form the theoretical basis for empirical testing in chapter 4, where system GMM is applied to evaluate which financial metrics most consistently predict semiconductor stock returns. The calculations for these metrics are presented in the Appendix.

4 Methodology

4.1 Data Estimation Method

This thesis uses the two-step system Generalized Method of Moments (GMM) estimator to achieve consistent estimates in the dynamic panel of semiconductor firm returns. GMM avoids strong distributional assumptions (Hansen 1982) and instead relies on moment conditions, where valid instruments are uncorrelated with the error term and correlated with endogenous regressors. This makes the method particularly suitable for panels with many firms and relatively few time periods, where endogeneity, firm heterogeneity, and feedback effects complicate estimation.

The model is specified in general form as:

$$y_{it} = \alpha + \delta y_{i,t-1} + X'_{i,t-1}\beta + \eta_i + \lambda_t + \epsilon_{it}$$

Where y_{it} is the dependent variable, $y_{i,t-1}$ its lag, $X'_{i,t-1}$ a vector of lagged explanatory variables, η_i the time-invariant firm-specific effect, and λ_t the time effect. All regressors are lagged by one period to avoid look-ahead bias and ensure consistency with information available at the start of year t .

First differences remove the firm-specific effect η_i , resulting in:

$$\Delta y_{it} = \delta \Delta y_{i,t-1} + \Delta X'_{i,t-1}\beta + \Delta \lambda_t + \Delta \epsilon_{it}$$

Arellano and Bond (1991) propose using lagged levels as instruments for the differenced equation, but when variables are highly persistent and the time dimension is short, these instruments may be weak and lead to biased estimates. To address this, Arellano and Bover (1995) and Blundell and Bond (1998) extend the estimator by adding the equation in levels, instrumented with lagged first differences. This system of equations improves efficiency under stationarity and mitigates identification problems caused by weak instruments.

To limit the risk of instrument proliferation, the instrument count is restricted to remain below the number of firms by employing a collapsed matrix (Roodman 2009). Standard errors are adjusted using Windmeijer's (2005) finite-sample correction to address downward bias in two-step GMM. In practice, this correction ensures more reliable inference for the financial metrics, reducing the likelihood of overstating significance.

Prior to estimation, variance inflation factor (VIF) tests are used to detect multicollinearity among financial metrics, guiding the choice of model specification that improve precision. Alternative specifications are estimated for robustness, and model validity is tested using the Arellano-Bond test for second-order autocorrelation and the Hansen J-test of over-identifying restrictions.

Compared with alternatives such as OLS or fixed effects, which suffer from bias in dynamic panels, and Fama-Macbeth regressions, which are less suited to short panels, System GMM provides a consistent and efficient estimator in the presence of endogenous regressors. Its application here is not only used to explain returns, but to identify financial metrics that can be translated into transparent investment formula for the semiconductor industry.

4.2 Empirical Model

The empirical model is designed to test the predictive power of firm fundamentals for semiconductor stock returns within a dynamic panel. Annual firm-level stock returns are modelled with lagged financial metrics, while accounting for firm and time effects. The general specification is

$$R_{it} = \alpha + \delta R_{i,t-1} + \beta_1 ROE_{i,t-1} + \beta_2 ROA_{i,t-1} + \beta_3 ROCE_{i,t-1} + \beta_4 OM_{i,t-1} + \beta_5 DE_{i,t-1} \\ + \beta_6 R\&D_{i,t-1} + \beta_7 RG_{i,t-1} + \eta_i + \lambda_t + \varepsilon_{it}$$

Where R_{it} is the total return for firm i in year t . All explanatory variables are lagged by one year to prevent look-ahead bias and to ensure consistency with the information available to investors at the start of year t . Firm fixed effects η_i capture unobserved heterogeneity across firms, while year dummies λ_t absorb market-wide shocks and semiconductor cycle effects. The

error term ε_{it} reflects firm-specific return shocks not explained by the fundamentals or fixed effects.

The inclusion of the lagged dependent variable $R_{i,t-1}$ captures return persistence, (momentum effect), while the financial ratios correspond to the theoretical determinants introduced in Chapter 3. Profitability is represented by ROE, ROA, and ROCE; efficiency by the operating margin; leverage by the debt-to-equity ratio; innovation by R&D intensity; and growth by revenue expansion.

The structure of the empirical model follows the objective of evaluating whether firm fundamentals contain predictive information for future semiconductor stock returns. Starting from a dynamic panel, the specification aims to capture behavioural and structural effects through lagged metrics and past performance. The lagged specification is to replicate a real-world scenario for the individual investor and avoid any look ahead bias. Additionally, firm level metrics are treated as predetermined when included in lagged form, meaning that last year's accounting information may reflect earlier shocks but is assumed not to respond to the current period return shock.

Since the purpose is return predictability rather than estimating an expected-return model, the specification does not include a separate term for the market's expected return. Market-wide movements are not firm-specific and therefore do not help to identify which fundamentals explain cross-firm differences in returns. The analysis instead isolates the predictive role of firm characteristics in the panel, while systematic market exposure is evaluated when the portfolios are constructed and tested with the CAPM.

4.3 Validity and Diagnostic Tests

To be certain about the validity of the System GMM estimates, a set of diagnostic tests is conducted. These tests evaluate whether the instruments satisfy the required conditions, including instrument validity and the absence of higher-order serial correlation. When the model passes these tests, the results can be interpreted with confidence and form a reliable foundation for the portfolio construction presented below.

4.3.1 Instrument Validity: Hansen J-test

The Hansen J-test assesses whether the instruments used in the Two-step System GMM estimation are valid, meaning that they are uncorrelated with the error term. The estimation relies on moment conditions of the form $E[z_i \varepsilon_i] = 0$, i.e. that the chosen instruments must be uncorrelated with the model's error term. Under the null hypothesis of valid instruments, the

Hansen statistics follows a chi-square distribution, where the degrees of freedom reflect the number of overidentifying restrictions (Hansen 1982).

The purpose of the test is to ensure that the dynamic panel specification is not distorted by endogeneity arising from firm-specific characteristics or persistent industry shocks. Since the model includes several potentially correlated variables and a lagged dependent variable, verifying instrument validity is necessary to ensure inference without bias.

4.3.2 Serial Correlation: Arellano-Bond Autocorrelation Tests

To measure serial correlation for the specification, the Arellano Bond tests is used. The model is expected to have a significant first order correlation due to the differencing process, while the no second order correlation is required for the moment conditions to hold. Under the null hypothesis, the differenced residuals are free from second order correlation, and the corresponding test statistic is asymptotically normally distributed (Arellano and Bond 1991).

Establishing this property is important because the validity of the instruments depends on the structure of the error term. Evidence of second order correlation would imply that $E[y_{i,t-2} \Delta \varepsilon_{it}] = 0$ no longer holds, since the lagged levels used as instruments would be correlated with the differenced errors. This would violate the moment condition of the estimator.

4.3.3 Joint Significance: Wald Test for Coefficients and Time Dummies

The Wald test evaluates whether a set of coefficients in the specification is jointly significant by testing the null hypothesis that the parameters in question are equal to zero. More precisely, the test examines whether the estimated vector differs from the zero vector once sampling variation is taken into account, and the resulting statistics follow a chi square distribution under the null hypothesis (Wald 1943). For the explanatory variables, the purpose of the test is to verify that the selected financial metrics contribute to the dynamic panels as a group. Since individual coefficients may be influenced by noise, the joint test provides a broader indication of whether the specification captures systematic variation in semiconductor stock returns rather than random uncorrelated effects.

The same reasoning applies to the time dummies included in the model. Testing their joint significance determines whether common shocks affecting the semiconductor industry across years contribute meaningfully to the variation in returns. A significant result in this case indicates that year specific effects, such as market wide cycles or technology driven fluctuations, should be included in the specification to avoid omitted variable bias.

4.4 Common Stock Selection

To translate the econometric results into practical investment strategies, two methods are applied. Both rely exclusively on information available at the time of selection to prevent look-ahead bias. Firms are ranked annually according to their financial metrics, based on Joel Greenblatt's Magic Formula (2010), with the top-ranked firms forming equally weighted portfolios.

4.4.1 Annual Rebalancing Strategy

At the start of each year t , firms are ranked using accounting data from year $t - 1$. The five firms with the highest values according to the selected metric enter an equally weighted portfolio, which is then held for a full year. At the beginning of year $t + 1$, the ranking is updated with new information, and a new portfolio is constructed. This process repeats annually, ensuring that portfolio composition adapts to evolving firm fundamentals and industry conditions.

Rebalancing means updating the portfolio so that its stocks reflect the firms that currently satisfy the selection rule based on the most recent annual accounting information. It allows the portfolio to adjust as market conditions and firm characteristics evolve. This approach recognises that the competitive market of semiconductors shifts over time and that firms that appear attractive in one year may be replaced by stronger performers in the next. Updating the portfolio annually ensures that the strategy can incorporate these changes and avoids excluding new firms that meet the selection criteria, which supports the objective of capturing the highest possible return from the available information.

4.4.2 Long-Term Buy and Hold Strategy

The second method forms a single portfolio at the beginning of the investment period. Firms are screened once based on year $t - 1$ metrics, and the top five firms are selected with equal weights. This portfolio is then held without rebalancing for six years. The strategy tests whether firms identified as fundamentally strong at a single point in time can continue to deliver superior returns over an extended horizon without active adjustment.

4.5 Measuring Excess Returns

To determine whether the portfolios generate returns beyond what can be explained by systematic risk, performance is tested within the Capital Asset Pricing Model (CAPM) framework. Jensen's alpha is estimated using time-series regression of portfolio excess returns

on market excess returns. All portfolio returns are expressed in excess of the risk-free rate, and heteroskedasticity-robust standard errors are reported.

The general specification is:

$$R_{p,t} - R_{f,t} = \alpha_p + \beta_{pM} (R_{M,t} - R_{f,t}) + \varepsilon_{p,t}$$

where $R_{p,t}$ is the return of portfolio p in month t , $R_{f,t}$ is the one-month Treasury bill rate, and $R_{M,t}$ is the market return. The coefficient β_{pM} captures systematic exposure to market, while intercept α_p measures abnormal performance unexplained by risk factors. In equilibrium, CAPM predicts $\alpha_p = 0$. A positive and statistically significant α_p indicates that the portfolio delivers excess returns.

5 Data

5.1 Sample Construction and Sources

The dataset consists of publicly listed firms classified under the Semiconductors and Semiconductor Equipment industry group (GICS 4530). Firms were identified using the GICS classifications maintained by MSCI and S&P Global. The sample includes integrated device manufacturers (IDMs), fabless firms, foundries, and equipment suppliers. After applying this classification, the final sample contains 50 firms with annual observations from 2006 to 2022 across North America, Europe, and Asia. Each firm is presented in Table A1 in the Appendix.

Financial information was collected from approximately 850 annual reports using Refinitiv Eikon. Accounting data originate from balance sheets, income statements, and cash flow statements and are reported in USD. The NASDAQ monthly return and the 10-year US Treasury yield (monthly) used in the CAPM regression were also obtained from Refinitiv Eikon. Stock prices are recorded annually in local currency. This frequency corresponds to firms' reporting cycles and follows common practice in empirical asset-pricing research (Asness, Frazzini, and Pedersen 2019, Ghazal et al. 2024, Gschwandtner and Hauser 2016, and Tsung-chun et al. 2019). Portfolio returns for Section 6.2 are collected from Refinitiv Eikon and reported in Tables A4.1–A4.3.

5.2 Descriptive Statistics

Table 5.1 shows large volatility in annual stock returns, which is consistent with the semiconductor industry's high demand and market volatility (Ehm and Ponsignon 2012). Profitability ratios such as ROE and ROCE display extreme values in the data, often caused by negative equity, one-off impairment, or accounting changes. These are typical in semiconductor

firms given their large amount of fixed assets and large depreciation costs. Scatter plots of returns against each metric are reported in Appendix A2.1 and suggest that ROA and revenue growth exhibit the clearest positive correlation with current returns, while most other variables show no stable unconditional relationship; the plots also highlight the extent of extreme observations.

Table 5.1: Descriptive statistics of financial metrics

Variables	Mean	Std. dev.	Min	Max
Return _t	10.00%	45.00%	−244.00%	194.00%
ROE _t	−17.05%	786.47%	−22,400.00%	292.70%
ROA _t	9.45%	13.54%	−57.59%	83.74%
ROCE _t	7.58%	191.85%	−5,502.09%	155.77%
Operating Margin _t	10.74%	130.24%	−3,617.83%	491.10%
DE _t	53.42%	129.72%	0.04%	2,032.34%
R&D Intensity _t	14.80%	8.21%	−34.17%	42.49%
Revenue Growth _t	9.00%	24.00%	−107%	122%
N.obs.	840			

Note: The total number of observations is 840. The sample size for Return (N = 787).

Operating margin differs across business models, which contributes to heterogeneity in the panel. Qiao and Lu (2024) classify semiconductor firms into distinct business model groups and show that fabless firms are less capital intensive, while foundries and OSATs are capital intensive and IDMs combine both labour intensive and capital-intensive activities. Leverage also varies, with fabless firm relying less on debt and IDMs maintaining higher long-term debt due to fabrication investments. R&D intensity generally lies between 10 and 20 percent of revenue, which is consistent with industry statistics reported by the NCSSES (2024).

In summary, the descriptive statistics reflect the heterogeneity of the semiconductor sector, with variations driven by business structure, investment cycles, and reporting effects. However, there is also extreme values present in the data that will require careful treatment before estimation.

5.2.1 Data Cleaning and Outlier Treatment

The initial dataset is an unbalanced panel of 850 observations. To limit the effect of reporting errors, negative equity episodes, and other accounting anomalies, financial ratios were winsorised using boundaries informed by industry characteristics and previous empirical studies. The aim is to exclude observations that do not reflect the underlying firm performance. In semiconductor firms, large changes in revenue are common, but profitability and capital ratios rarely reach the extreme levels presented in the dataset. Removing them preserves the economic content of the data without modifying it for the estimation method.

The winsorisation of financial ratios follows common practice in the accounting and asset pricing literature and is applied to limit the influence of extreme observations. Extremely large ratios, as mentioned, are often driven by non-recurring accounting items, impairments and short-term fluctuations that are unlikely to reflect sustainable operating performance. Prior empirical research therefore applies winsorisation to reduce the influence of such distortions, including Chen et al. (2018), Gschwandtner and Hauser (2016), and Novy-Marx (2013). In line with this practice, all metrics in this study are two sided winsorised at the first and ninety ninth percentiles. This will retain all firm year observation but replace values at end of the tails with corresponding percentile cut offs, thereby reducing the effects of extreme outlier while preserving the panel.

The winsorised data set is summarised in Table 5.2. All metrics are lagged so that they reflect the information available at the end of year t and are used to explain the stock return in year $t + 1$. The number of observations is lower than in the raw data because forming one period lags removes the first year of observations for each firm, meaning that returns are observed from 2007-2022, and the lagged metrics from 2006-2021. The panel is unbalanced due to missing values in some financial reports, and the metrics still vary substantially across firms even after winsorisation. ROE is more volatile than ROA, and Debt to Equity retains very large values. Operating Margin and Revenue Growth also differ widely across firms and across years, which points to meaningful differences in firm characteristics and cyclical variation over time. This is the sample used in the System GMM estimation. Figure A2.2, in the Appendix, presents the lagged and winsorised metrics against returns and shows the strongest positive correlation for revenue growth. ROA and ROE display weaker positive correlations, while the remaining metrics show little visible correlation with returns.

Table 5.2: Descriptive statistics of winsorised and lagged metrics

Variables	Mean	Std dev	Min	Max
ROE _{t-1}	12.77%	24.43%	-104.22%	78.83%
ROA _{t-1}	8.93%	12.69%	-31.89%	39.49%
ROCE _{t-1}	13.36%	17.19%	-23.77%	93.83%
Operating Margin _{t-1}	13.34%	20.41%	-60.90%	117.29%
DE _{t-1}	46.28%	62.83%	0.21%	400.58%
R&D Intensity _{t-1}	15.00%	7.44%	-9.29%	34.33%
Revenue Growth _{t-1}	9.00%	23.00%	-55.00%	79.00%
N.obs.	790			

Note: The sample size varies by variable due to data availability. Specifically, DE (N = 750), R&D Intensity (N = 760), and Revenue Growth (N = 739) reflect smaller sub-samples.

6 Results and Discussion

6.1 System GMM estimation

Before presenting the System GMM results, it is necessary to examine the correlation between the lagged firm fundamentals and stock returns in more detail. Figure A2.2 in the appendix plots each lagged financial metric against annual returns. Most variables, including debt to equity, operating margin and R&D intensity, show no clear relationship with future returns. ROA and revenue growth indicate some correlation.

Variance inflation factor tests reported in Table 6.1 indicate moderate multicollinearity among the profitability measures. ROA and ROE exhibit the highest VIF values, suggesting that they capture overlapping information, while ROCE and operating margin show smaller but still nonzero multicollinearity. In contrast, leverage, revenue growth and R&D intensity display low VIF values and does not raise concerns. These results imply that profitability should not be represented by several closely related ratios within the same specification. Nevertheless, profitability measures are in a full model using fixed effects, and separately in GMM estimation to ensure that the main conclusion is transparent and correctly motivated.

Table 6.1: Variance Inflation Factor

Variable	VIF
ROE (lagged)	5.79
ROA (lagged)	7.39
ROCE (lagged)	2.08
Operating Margin (lagged)	2.02
Debt-to-Equity (lagged)	1.16
R&D Intensity (lagged)	1.24
Revenue Growth (lagged)	1.30

Given the VIF values, estimating a full fixed effects model with all the metrics may cause issues for the individual coefficients, particularly because ROA and ROE capture overlapping information. The fixed effects specification (A3.1) is included as a baseline and is estimated as a full model that includes the metrics but excludes return persistence. None of the coefficients are statistically significant, which shows that the metrics do not jointly explain returns once

time invariant characteristics are removed. This result should not be interpreted as proof that firm level information is irrelevant, since it is a strict joint test where correlated profitability measures compete for the same variation and can weaken inference.

For the main analysis, the goal is not to maximise fit in one full specification, but to identify which fundamentals provide reliable predictive information once endogeneity and firm heterogeneity are considered. System GMM is therefore estimated in simple and transparent specifications that (i) avoid severe multicollinearity, (ii) limit instrument proliferation, and (iii) remain easy to interpret as screening rules. Each metric is tested on its own and is only retained if it satisfies the diagnostic requirements in Section 4.3, i.e. the Hansen test does not reject, the AR(2) test does not reject, and the instrument count remains limited. This also matches the practical aim of the thesis, since including many closely related ratios would complicate the investment rule without adding clear information.

Following the diagnostic criteria in Section 4.3, a specification is only treated as valid if it passes the Hansen test, does not reject the AR(2) condition, and keeps the instrument count limited to avoid instrument proliferation. Appendix A3.2 to A3.6 therefore reports alternative models estimated with different lagged financial metrics to evaluate which variables meet these requirements. The ROE model (A3.2) yields a positive and statistically significant coefficient, but it fails the Arellano Bond test for second order autocorrelation, which violates a necessary condition for valid GMM inference. The ROCE model (A3.3) is statistically significant at the 10 percent level, but it also fails the AR(2) diagnostic. Operating margin (A3.4) is not statistically significant, and the AR(2) test rejects. The revenue growth model (A3.5) is statistically significant, but it fails the Hansen test and rejects the AR(2) condition. Debt-to-Equity has significant return persistence; however, it fails on the Hansen and AR(2) condition (A3.6) and the same goes for R&D intensity model (A3.7). In contrast, the preferred ROA specification reported in Table 6.2 is selected because it satisfies the standard diagnostic requirements while being statistically significant.

In Table 6.2 the specification is presented. The estimated coefficient is positive and statistically significant, indicating that firms with higher returns on their assets earn higher stock returns. In contrast, the lagged return is small and statistically insignificant, which suggest that return persistence on an annual basis does not influence return once the firm fundamentals are considered. The Hansen test does not reject the null of valid instruments, and the Arellano Bond tests indicate first order but not second order autocorrelation, which is consistent with the requirements of the estimator. The Wald tests confirm joint significance of the explanatory

variables and the time effects, which means that both firm level profitability and time contribute to explaining stock returns.

Table 6.2: System GMM - ROA

Variable		<i>Return</i>
Return (t-1)		0.037
		(0.052)
ROA (t-1)		0.429***
		(0.114)
Obs.		787
Groups		50
Time dimension		11-16
Nr of Obs. Used		1 418
Hansen test	$\chi^2(2)$	0.886
Autocorrelation test (1)		-3.881***
Autocorrelation test (2)		-1.786
Wald test for coefficients	$\chi^2(2)$	41.835***
Wald test for time dummies	$\chi^2(14)$	660.364***

Note: *** zero point one percent significance level, ** one percent significance level, * five percent significance level, . ten percent significance level.

From an economic perspective, the results were expected given the structure of the semiconductor industry. Production relies on expensive fabrication plants, equipment, and infrastructure, making efficient use of assets a determinant of financial performance. ROA captures this directly, whereas ROE is more sensitive to capital structure decisions, which varies across firm types. This makes ROA more stable, informative and suitable measure for creating a metric-based investment formula.

In regards to prior research, the findings align with the evidence that profitability is an important determinant of stock returns (Asness, Frazzini, and Pedersen 2019; Gschwandtner and Hauser 2016). However, this study isolates ROA as the dominant predictor within the semiconductor industry, rather than broader quality measures suggested in cross sector analyses. Based on this result, the investment rule is deliberately reduced to a single metric

$$\text{Return on Assets} = \frac{\text{Income before tax}}{\text{Average Total Assets}}$$

If average total assets are unavailable, total assets from the balance sheet of the corresponding year are used instead. This provides a transparent and replicable selection rule. Combined with the portfolio methodologies presented above, the strategy can adapt to changes in firm performance and industry structure.

6.2 Performance Evaluation of ROA-based Portfolio Strategies

This section evaluates the practical implications of implementing the ROA-based selection rule as a transparent investment strategy. Performance is compared across three ROA-based strategies and two benchmarks, and outcomes are reported as compounded and annualised returns. As shown in Section 6.1, ROA is identified as the robust profitability metric, but the profit persistence literature suggests that unusually high profitability can reflect either sustainable competitive advantage or transitory shocks that later mean-revert, and that persistence may differ in the upper tail (Canarella, Miller, and Nourayi 2013). This motivates evaluating two ROA caps (30% and 50%) as a sensitivity check to test whether portfolio performance is driven by extreme high-ROA observations. The purpose is to assess whether the profitability signal identified in the econometric analysis translates into a tangible strategy when applied as an investment formula.

Figure 6.1 shows cumulative performance between April 2018 and April 2024. All strategies generate large returns during this period, but the 30% cap ROA-based portfolio delivers the strongest growth overall. Table 6.3 summarises annualised and total returns. The rebalanced ROA-30% strategy achieves the highest annualised return (27.12 percent), exceeding both NASDAQ (19.20 percent) and the S&P Semiconductor Index (26.78 percent). The buy-and-hold strategy also performs strongly (24.76 percent), but it underperforms the rebalanced ROA-30% strategy, suggesting that updating the portfolio matters.

Figure 6.1: Compounded Returns of Portfolio Strategies, 2018-2024

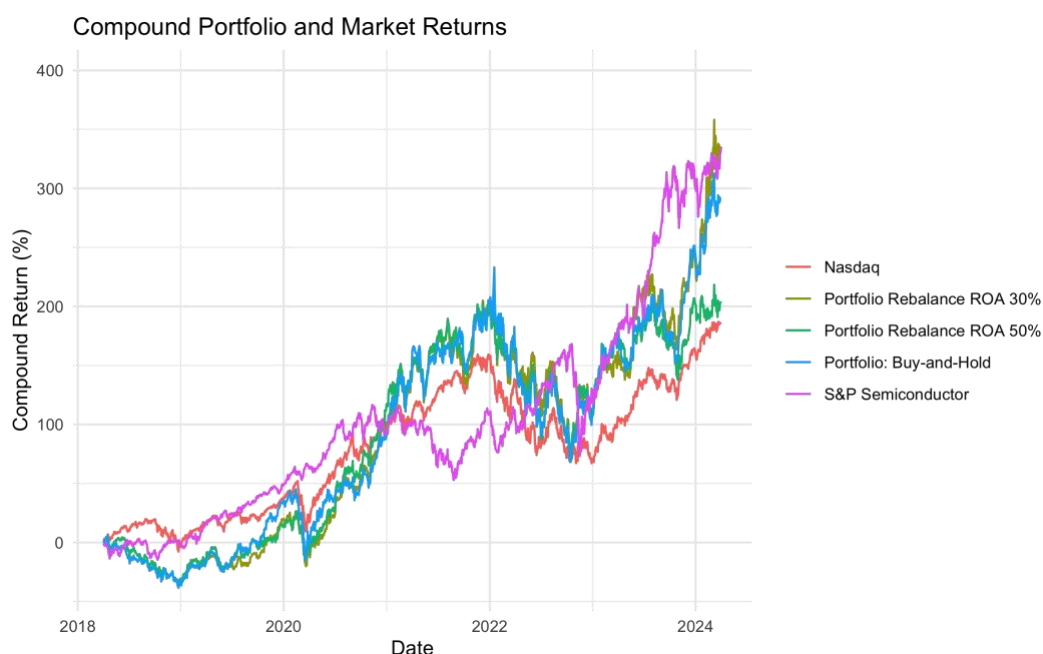


Table 6.3: Portfolio Performance, 2018-2024

Portfolio's	Annualized Return	Total return
NASDAQ	19.20%	186.24%
Rebalance (50%)	19.81%	204.22%
Rebalance (30%)	27.12%	335.35%
Buy-and-Hold	24.76%	289.89%
S&P Semiconductor	26.78%	334.84%

The difference between the 30 percent and 50 percent ROA caps is informative. ROA is an asset-based measure and can become extreme. In a capital-intensive industry such as semiconductors, unusually high ROA values are more likely to reflect temporary conditions than persistent operating efficiency. Imposing a 30 percent cap is a conservative way to focus the signal on sustained profitability rather than short-lived spikes seemingly captured by the 50 percent portfolio. The outperformance of the 30 percent cap relative to 50 percent is consistent with the interpretation that very high ROA observations add noise rather than durable information for portfolio selection.

The rebalanced strategies outperformed the buy-and-hold portfolio because they include updated metrics which changes over time. This is particularly relevant in an industry where profitability evolves with product cycles, capacity conditions, and technological shifts. Annual rebalancing allows firms with improving profitability to enter the portfolio and firms with weakening profitability to exit. The difference in performance between rebalanced and static allocations supports the idea that the profitability signal is not purely a one-off effect but remains relevant when applied repeatedly through time.

Benchmark comparisons provide additional context. The gap relative to NASDAQ is large, whereas the difference relative to the S&P Semiconductor Index is smaller. This indicates that a large share of realised return is driven by broad sector dynamics during the sample period. The semiconductor benchmark is therefore reported as a sector reference rather than as the main object of inference since it captures industry-wide movements that are not the focus of the screening rule. The fact that the ROA-based strategy is close to the sector index in aggregate return suggest that a large part of the variation is explained by common semiconductor conditions over this period. At the same time, the ROA-30% strategy marginally exceeds the sector benchmark in annualised terms, which represents the incremental return the rule generates.

From the perspective of the adaptive market hypothesis, this pattern is plausible. If market efficiency varies over time and information is incorporated with frictions, profitability signals may retain predictability for some periods, while the sector cycle remains the dominant driver of returns.

The results are also consistent with Andersson (2024), who reports strong performance for ROA-based strategies in the renewable energy sector. In conclusion, these results suggest that asset-based profitability may be informative in capital-intensive industries.

6.3 Risk-Adjusted Performance Evaluation

To examine whether the ROA-30% portfolio generates excess returns beyond those explained by systematic market risk, the CAPM is estimated using monthly excess returns over the sample period. The dependent variable is the portfolio's monthly excess return, and the independent variable is the NASDAQ excess return.

Figure A4 in the Appendix plots the portfolio's return against the market, with the fitted regression line illustrating the estimated beta. Table 6.4 reports the regression results. The estimated beta is 1.117, indicating that the portfolio carries higher systematic risk than the

market: on average, returns increase or decrease by 1.117 percent for every 1 percent movement in the market. Jensen's alpha is positive but statistically insignificant, implying that after adjusting for market risk, the portfolio's excess return cannot be distinguished from zero.

Table 6.4: CAPM Regression Result

Variable	Portfolio returns
Jensen's Alpha	0.0063 (0.0068)
Market Return	1.117*** (0.113)
R ²	0.5647
Obs	73

Note: *** one percent significance level, ** five percent significance level, * ten percent significance level. Robust standard errors used to obtain t-values.

This outcome suggests that the superior returns observed for the ROA-based portfolio are consistent with higher exposure to market risk, rather than by an independent effect of the selection rule. This aligns with the CAPM prediction that only systematic risk is priced in equilibrium, while firm-specific characteristics such as high ROA should not deliver persistent abnormal returns once risk is accounted for.

Overall, the CAPM results indicate that the ROA-30% portfolio does not deliver statistically significant excess returns once systematic risk is considered. Its superior performance is therefore primarily attributed to larger risk and market exposure. Nevertheless, the positive alpha is consistent with the AMH view that strategies based on persistent fundamentals may vary in effectiveness across market conditions, even if these are not stable across all periods.

7 Conclusion

This thesis investigated which firm-specific financial metrics explain semiconductor stock returns and whether a fundamentals-based formula can outperform the market benchmarks. System GMM estimation identified return on asset (ROA) as the most robust predictor of future returns, and portfolio tests showed that profitability-based screening can deliver strong performance, particularly when updated annually.

The CAPM analysis indicated that the superior returns of the ROA-based strategy are largely explained by higher market exposure, with no statistically significant alpha. Still, the presence of a positive alpha, although inconclusive, suggest that market may occasionally underreact to persistent fundamentals.

The broader implication is that profitability-based strategies may be especially valuable when applied to industries on the verge of structural growth. In this study, investing in semiconductors already outperformed the NASDAQ, and further filtering by ROA added incremental gains. Identifying the next emerging industry could generate substantial returns.

Future research could extend the methodology to other industries, test alternative weighting schemes and incorporate additional econometric method to refine the model. Additionally, the AMH provides a useful perspective for understanding when and why fundamental strategies succeed and should be considered a standard theory when evaluating market efficiency.

Lastly, the results show that, once unobserved firm effects and endogeneity are accounted for, ROA is the only robust fundamental predictor of future stock returns, while momentum/persistence effects at the annual frequency are not statistically significant. This narrows the interpretation of “quality” in this sector to asset efficiency and supports a simple screening formula that reduces the number of firms investors needs to evaluate.

8 References

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9 Appendix

A1 Sample Composition

Table A1: Firms 2006-2022

Firm	Ticker	Type
NVIDIA Corporation	NVDA	Fabless
Broadcom Inc.	AVGO	Fabless
Taiwan Semiconductor Manufacturing Co. (TSMC)	TSM	Foundry
ASML Holding N.V.	ASML	Equipment
Advanced Micro Devices, Inc.	AMD	Fabless
QUALCOMM Incorporated	QCOM	Fabless
Texas Instruments Incorporated	TXN	IDM
Applied Materials, Inc.	AMAT	Equipment
Analog Devices, Inc.	ADI	IDM
Lam Research Corporation	LRCX	Equipment
SK Hynix Inc.	000660.KS	IDM
Micron Technology, Inc.	MU	IDM
Intel Corporation	INTC	IDM
MediaTek Inc.	2454.TW	Fabless
Semiconductor Manufacturing International Corp. (SMIC)	0981.HK	Foundry
Marvell Technology, Inc.	MRVL	Fabless
NXP Semiconductors N.V.	NXPI	Fabless
Monolithic Power Systems, Inc.	MPWR	Fabless

Microchip Technology Incorporated	MCHP	IDM
STMicroelectronics N.V.	STM	IDM
Cohu, Inc.	COHU	Equipment
United Microelectronics Corporation	UMC	Foundry
ON Semiconductor Corporation	ON	IDM
Wolfspeed, Inc.	WOLF	IDM
Ultra Clean Holdings, Inc.	UCTT	Equipment
Novatek Microelectronics Corp.	3034.TW	Fabless
Skyworks Solutions, Inc.	SWKS	Fabless
BE Semiconductor Industries N.V.	BESI.AS	Equipment
Lattice Semiconductor Corporation	LSCC	Fabless
Qorvo, Inc.	QRVO	Fabless
Onto Innovation	ONTO	Equipment
Nova Ltd.	NVMI	Equipment
Vanguard International Semiconductor Corp.	5347.TWO	Foundry
Cirrus Logic, Inc.	CRUS	Fabless
Alpha & Omega Semiconductor Ltd.	AOSL	Fabless
Tower Semiconductor Ltd.	TSEM	Foundry
Silicon Laboratories Inc.	SLAB	Fabless
Camtek Ltd.	CAMT	Equipment
Semtech Corporation	SMTC	Fabless

Power Integrations, Inc.	POWI	Fabless
ASMPT Ltd. (ASM Pacific Technology)	0522.HK	Equipment
Melexis NV	MLXS.BR	Fabless
FormFactor, Inc.	FORM	Equipment
Nordic Semiconductor ASA	NOD.OL	Fabless
Vishay Intertechnology, Inc.	VSH	IDM
Silicon Motion Technology Corp.	SIMO	Fabless
Kulicke and Soffa Industries, Inc.	KLIC	Equipment
Aixtron SE	AIXGn.DE	Equipment
Axcelis Technologies, Inc.	ACLS	Equipment
Veeco Instruments Inc.	VECO	Equipment

Note: All firms included in the sample.

A2. Independent Variables and Scatterplots

$$ROE = \frac{\text{Income before tax}}{\text{Average Shareholders' Equity}} \quad (A2.1)$$

$$ROA = \frac{\text{Income before tax}}{\text{Average Total Assets}} \quad (A2.2)$$

$$ROCE = \frac{EBIT}{\text{Capital Employed}} \quad (A2.3)$$

$$\text{Operating Margin} = \frac{\text{Operating Income}}{\text{Net Sales}} \quad (A2.4)$$

$$D/E = \frac{\text{Total Debt}}{\text{Total Equity}} \quad (A2.5)$$

$$R\&D \text{ Intensity} = \frac{\text{R\&D Expenditures}}{\text{Total Revenue}} \quad (A2.6)$$

$$\text{Revenue Growth} = \ln\left(\frac{\text{Total Revenue}_t}{\text{Total Revenue}_{t-1}}\right) \quad (A2.7)$$

Figure A2.1: Scatterplot of Financial Metrics and Return

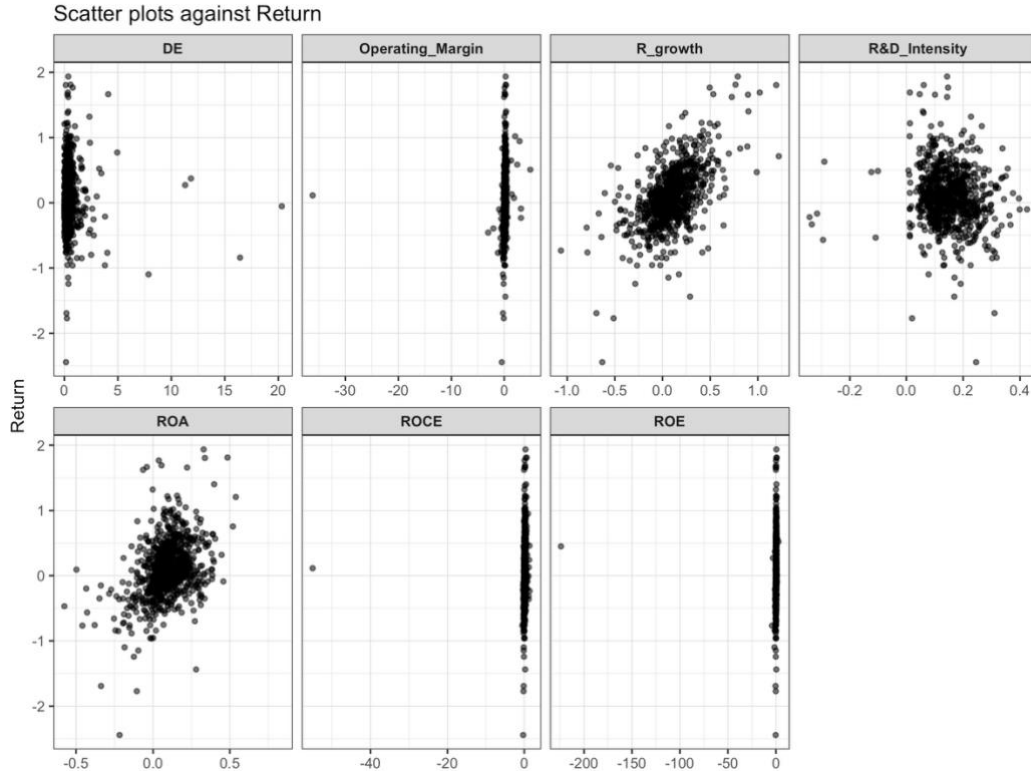
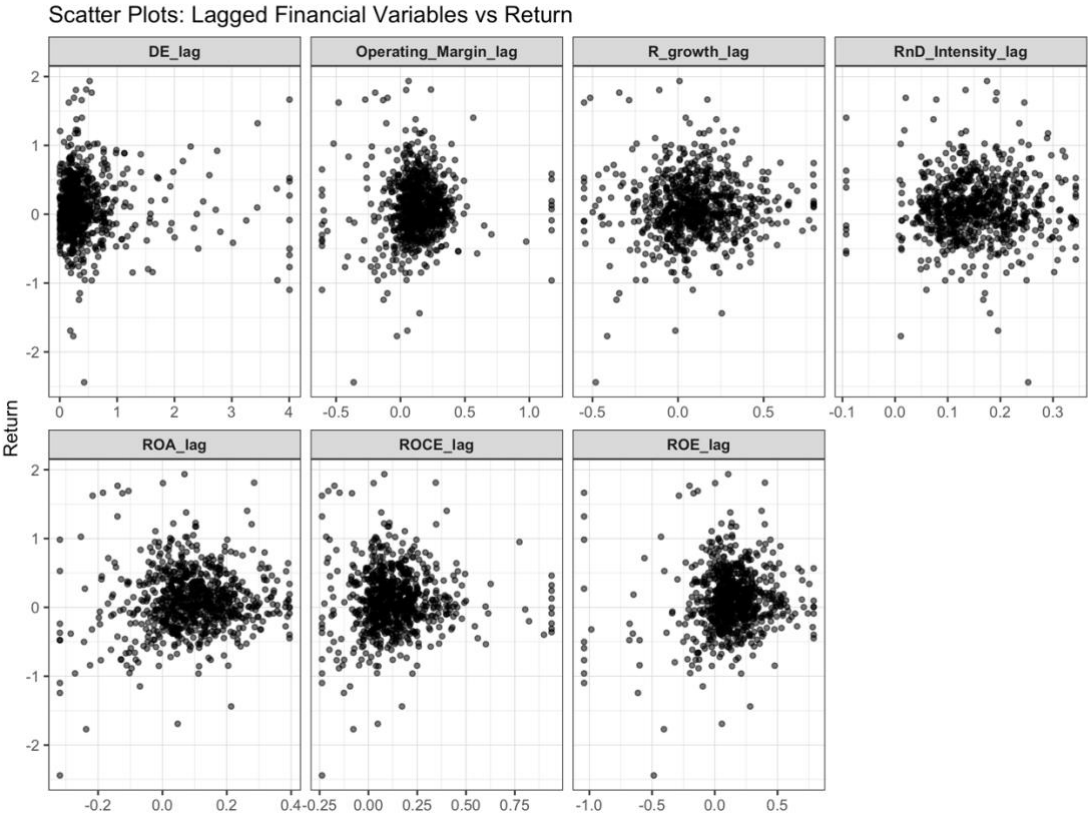


Figure A2.2: Scatterplot of Lagged Winsorised Variables



A3. Model Specifications

Table A3.1: Fixed Effects Regression Result – Full Model

Variable	Return
ROE(t-1)	0.051 (0.148)
ROA(t-1)	0.574 (0.383)
ROCE(t-1)	-0.013 (0.285)
Operating Margin(t-1)	-0.159 (0.124)
DE(t-1)	0.018 (0.030)
R&D Intensity(t-1)	-0.196 (0.411)
Revenue Growth(t-1)	0.166 (0.086)
Obs.	676
R ²	0.038
R ² Adj	-0.070

Note: *** one percent significance level, ** five percent significance level, * ten percent significance level. Robust standard errors used to obtain t-values.

Table A3.3: System GMM Estimation – ROE

Variable		<i>Return</i>
Return (t-1)		0.036
		(0.056)
ROE (t-1)		0.262***
		(0.065)
Obs.		784
Groups		50
Time dimension		11-16
Nr of Obs. Used		1 409
Hansen test	$\chi^2(2)$	0.876
Autocorrelation test (1)		-4.513***
Autocorrelation test (2)		-2.213*
Wald test for coefficients	$\chi^2(2)$	33.530***
Wald test for time dummies	$\chi^2(14)$	601.978***

Note: *** zero point one percent significance level, ** one percent significance level,
 * five percent significance level, . ten percent significance level.

Table A3.3: System GMM Estimation – ROCE

Variable		<i>Return</i>
Return (t-1)		0.036
		(0.056)
ROCE (t-1)		0.184*
		(0.080)
Obs.		784
Groups		50
Time dimension		11-16
Nr of Obs. Used		1 414
Hansen test	$\chi^2(2)$	0.397
Autocorrelation test (1)		-5.438***
Autocorrelation test (2)		-2.535*
Wald test for coefficients	$\chi^2(2)$	7.320*
Wald test for time dummies	$\chi^2(14)$	584.746***

Note: *** zero point one percent significance level, ** one percent significance level,

* five percent significance level, . ten percent significance level.

Table A3.4: System GMM Estimation –Operating Margin

Variable		<i>Return</i>
Return (t-1)		0.009
		(0.066)
Operating Margin (t-1)		0.272
		(0.174)
Obs.		786
Groups		50
Time dimension		11-16
Nr of Obs. Used		1 385
Hansen test	$\chi^2(2)$	0.060
Autocorrelation test (1)		-4.177***
Autocorrelation test (2)		-2.913**
Wald test for coefficients	$\chi^2(2)$	11.956**
Wald test for time dummies	$\chi^2(14)$	498.594***

Note: *** zero point one percent significance level, ** one percent significance level,
 * five percent significance level, . ten percent significance level.

Table A3.5: System GMM Estimation – Revenue Growth

Variable		<i>Return</i>
Return (t-1)		0.106
		(0.082)
Revenue Growth (t-1)		0.260*
		(0.105)
Obs.		739
Groups		50
Time dimension		11-16
Nr of Obs. Used		1 326
Hansen test	$\chi^2(2)$	13.252***
Autocorrelation test (1)		-3.272***
Autocorrelation test (2)		-2.389*
Wald test for coefficients	$\chi^2(2)$	26.988**
Wald test for time dummies	$\chi^2(13)$	326.797***

Note: *** zero point one percent significance level, ** one percent significance level,

* five percent significance level, . ten percent significance level.

Table A3.6: System GMM Estimation – Debt-to-Equity

Variable		<i>Return</i>
Return (t-1)		0.134*
		(0.067)
DE (t-1)		-0.029
		(0.024)
Obs.		746
Groups		50
Time dimension		11-16
Nr of Obs. Used		1 308
Hansen test	$\chi^2(2)$	0.964
Autocorrelation test (1)		-6.613***
Autocorrelation test (2)		-2.592**
Wald test for coefficients	$\chi^2(2)$	6.140*
Wald test for time dummies	$\chi^2(13)$	560.736***

Note: *** zero point one percent significance level, ** one percent significance level,
 * five percent significance level, . ten percent significance level.

Table A3.7: System GMM Estimation – R&D Intensity

Variable		<i>Return</i>
Return (t-1)		0.039
		(0.059)
DE (t-1)		-0.089
		(0.147)
Obs.		756
Groups		50
Time dimension		11-16
Nr of Obs. Used		1 366
Hansen test	$\chi^2(2)$	1.865
Autocorrelation test (1)		-3.978***
Autocorrelation test (2)		-1.764**
Wald test for coefficients	$\chi^2(2)$	1.556
Wald test for time dummies	$\chi^2(13)$	557.805***

Note: *** zero point one percent significance level, ** one percent significance level,
 * five percent significance level, . ten percent significance level.

A4 Portfolio Evaluation

Table A4.1: Buy-and-Hold Performance (2018-2024)

Period 1: 2018-04-03 – 2019-04-01	Annual return	Weights
Applied Materials, Inc. (AMAT)	-22.31%	20%
BE Semiconductor Industries N.V. (BESI.AS)	-41.68%	20%
Cirrus Logic, Inc. (CRUS)	5.16%	20%
Nova Ltd. (NVMI)	-3.08%	20%
Skyworks Solutions, Inc. (SWKS)	-13.17%	20%
Portfolio return	-15.02%	
Period 2: 2019-04-02 – 2020-04-01		
Applied Materials, Inc. (AMAT)	5.25%	20%
BE Semiconductor Industries N.V. (BESI.AS)	10.16%	20%
Cirrus Logic, Inc. (CRUS)	49.04%	20%
Nova Ltd. (NVMI)	24.89%	20%
Skyworks Solutions, Inc. (SWKS)	-1.67%	20%
Portfolio return	17.54%	
Period 3: 2020-04-02 – 2021-04-01		
Applied Materials, Inc. (AMAT)	226.84%	20%
BE Semiconductor Industries N.V. (BESI.AS)	173.00%	20%
Cirrus Logic, Inc. (CRUS)	44.81%	20%
Nova Ltd. (NVMI)	198.72%	20%
Skyworks Solutions, Inc. (SWKS)	127.65%	20%
Portfolio return	154.20%	
Period 4: 2021-04-02 – 2022-04-01		
Applied Materials, Inc. (AMAT)	-9.93%	20%
BE Semiconductor Industries N.V. (BESI.AS)	3.10%	20%
Cirrus Logic, Inc. (CRUS)	-4.14%	20%
Nova Ltd. (NVMI)	12.04%	20%

Skyworks Solutions, Inc. (SWKS)	-29.56%	20%
Portfolio return	-5.70%	

Period 5: 2022-04-02 – 2023-04-01

Applied Materials, Inc. (AMAT)	-3.62%	20%
BE Semiconductor Industries N.V. (BESI.AS)	5.53%	20%
Cirrus Logic, Inc. (CRUS)	30.54%	20%
Nova Ltd. (NVMI)	-2.52%	20%
Skyworks Solutions, Inc. (SWKS)	-10.72%	20%
Portfolio return	3.84%	

Period 6: 2023-04-02 – 2024-04-01

Applied Materials, Inc. (AMAT)	70.62%	20%
BE Semiconductor Industries N.V. (BESI.AS)	76.75%	20%
Cirrus Logic, Inc. (CRUS)	-16.13%	20%
Nova Ltd. (NVMI)	71.21%	20%
Skyworks Solutions, Inc. (SWKS)	-9.34%	20%
Portfolio return	38.62%	

Table A4.2: ROA 50% Performance (2018-2024)

Period 1: 2018-04-03 – 2019-04-01	Annual return	Weights
Melexis NV (MLXS.BR)	-27.53%	20%
Texas Instruments Incorporated (TXN)	-13.17%	20%
SK Hynix Inc. (000660.KS)	-45.64%	20%
Skyworks Solutions, Inc. (SWKS)	-13.17%	20%
BE Semiconductor Industries N.V. (BESI.AS)	-41.68%	20%
Portfolio return	-28.24%	
Period 2: 2019-04-02 – 2020-04-01		
SK Hynix Inc. (000660.KS)	7.40%	20%
Texas Instruments Incorporated (TXN)	5.16%	20%
Micron Technology, Inc. (MU)	-9.74%	20%
Melexis NV (MLXS.BR)	-27.53%	20%
NVIDIA Corporation (NVDA)	13.45%	20%
Portfolio return	-2.25%	
Period 3: 2020-04-02 – 2021-04-01		
Texas Instruments Incorporated (TXN)	42.74%	20%
Power Integrations, Inc. (POWI)	102.41%	20%
NVIDIA Corporation (NVDA)	185.08%	20%
QUALCOMM Incorporated (QCOM)	117.51%	20%
Vanguard International Semiconductor Corp. (3034.TW)	169.29%	20%
Portfolio return	123.41%	
Period 4: 2021-04-02 – 2022-04-01		
Texas Instruments Incorporated (TXN)	-3.13%	20%
Vanguard International Semiconductor Corp. (3034.TW)	-5.39%	20%
Taiwan Semiconductor Manufacturing Co., Ltd. (TSM)	-11.61%	20%
Applied Materials, Inc. (AMAT)	-10.03%	20%
NVIDIA Corporation (NVDA)	-6.08%	20%

Portfolio return	-7.25%	
Period 5: 2022-04-02 – 2023-04-01		
Texas Instruments Incorporated (TXN)	-2.52%	20%
Melexis NV (MLXS.BR)	-13.17%	20%
Advanced Micro Devices, Inc. (AMD)	-30.07%	20%
Kulicke and Soffa Industries, Inc. (KLIC)	-29.74%	20%
BE Semiconductor Industries N.V. (BESI.AS)	-41.68%	20%
Portfolio return	-23.44%	
Period 6: 2023-04-02 – 2024-04-01		
Melexis NV (MLXS.BR)	43.08%	20%
Alpha and Omega Semiconductor Limited (AOSL)	11.73%	20%
Texas Instruments Incorporated (TXN)	6.47%	20%
QUALCOMM Incorporated (QCOM)	19.28%	20%
Lam Research Corporation (LRCX)	21.85%	20%
Portfolio return	20.48%	

Table A4.3: ROA 30% Performance (2018-2024)

Period 1: 2018-04-03 – 2019-04-01	Annual return Weights	
Applied Materials, Inc. (AMAT)	-22.31%	20%
BE Semiconductor Industries N.V. (BESI.AS)	-41.68%	20%
Cirrus Logic, Inc. (CRUS)	5.16%	20%
Nova Ltd. (NVMI)	-3.08%	20%
Skyworks Solutions, Inc. (SWKS)	-13.17%	20%
Portfolio return	-15.02%	
Period 2: 2019-04-02 – 2020-04-01		
Vanguard International Semiconductor Corp. (5347.TWO)	-12.46%	20%
Applied Materials, Inc. (AMAT)	5.25%	20%
Lam Research Corporation (LRCX)	20.82%	20%
Nova Ltd. (NVMI)	24.89%	20%
Skyworks Solutions, Inc. (SWKS)	-1.67%	20%
Portfolio return	7.366%	
Period 3: 2020-04-02 – 2021-04-01		
Novatek Microelectronics Corp. (3034.TW)	238.57%	20%
Vanguard International Semiconductor Corp. (5347.TWO)	90.62%	20%
Lam Research Corporation (LRCX)	186.26%	20%
Qualcomm Incorporated (QCOM)	109.09%	20%
Skyworks Solutions, Inc. (SWKS)	127.65%	20%
Portfolio return	158.44%	
Period 4: 2021-04-02 – 2022-04-01		
Novatek Microelectronics Corp. (3034.TW)	-25.61%	20%
Applied Materials, Inc. (AMAT)	-9.93%	20%
Lam Research Corporation (LRCX)	-17.74%	20%
NVIDIA Corporation (NVDA)	92.90%	20%
Taiwan Semiconductor Manufacturing Co., Ltd. (TSM)	-17.57%	20%

Portfolio return	4.410%	
Period 5: 2022-04-02 – 2023-04-01		
Applied Materials, Inc. (AMAT)	-3.62%	20%
BE Semiconductor Industries N.V. (BESI.AS)	5.53%	20%
Lam Research Corporation (LRCX)	0.90%	20%
Qualcomm Incorporated (QCOM)	-13.30%	20%
Silicon Motion Technology Corp. (SIMO)	-0.61%	20%
Portfolio return	-2.22%	
Period 6: 2023-04-02 – 2024-04-01		
Applied Materials, Inc. (AMAT)	71.71%	20%
Kulicke and Soffa Industries, Inc. (KLIC)	-2.90%	20%
Monolithic Power Systems, Inc. (MPWR)	36.10%	20%
NVIDIA Corporation (NVDA)	227.73%	20%
Taiwan Semiconductor Manufacturing Co., Ltd. (TSM)	53.18%	20%
Portfolio return	77.164%	

Figure A4: CAPM Regression. ROA portfolio and NASDAQ Excess Market Returns

